Project Design Phase-II
Data Flow Diagram & User Stories

Date	28 October 2026
Team ID	xxxxxx
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	4 Marks

Project Overview & Scope:

Program Name: Data Analytics with Tableau

Project Context: Health360 analyzes 5,000 patient records across clinical vitals (blood pressure, glucose, cholesterol, BMI) and lifestyle factors (exercise, smoking, alcohol) to deliver real-time risk stratification.

Technology Stack: Python (Pandas/Flask), Tableau Desktop / Public, Tableau JavaScript API v2, HTML5/CSS3.

Deployment Host: Python Flask web server running locally on http://127.0.0.1:5001.

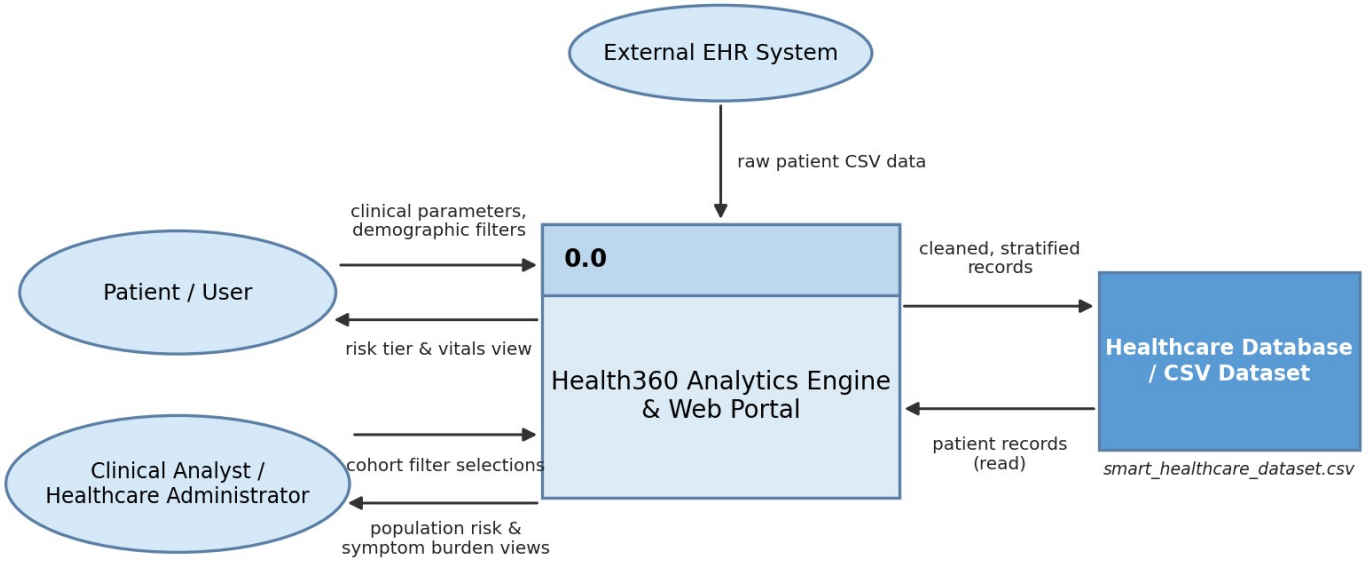
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

DFD Level 0 (Context Diagram) – Health360



DFD Level 1 (Decomposed Processes):

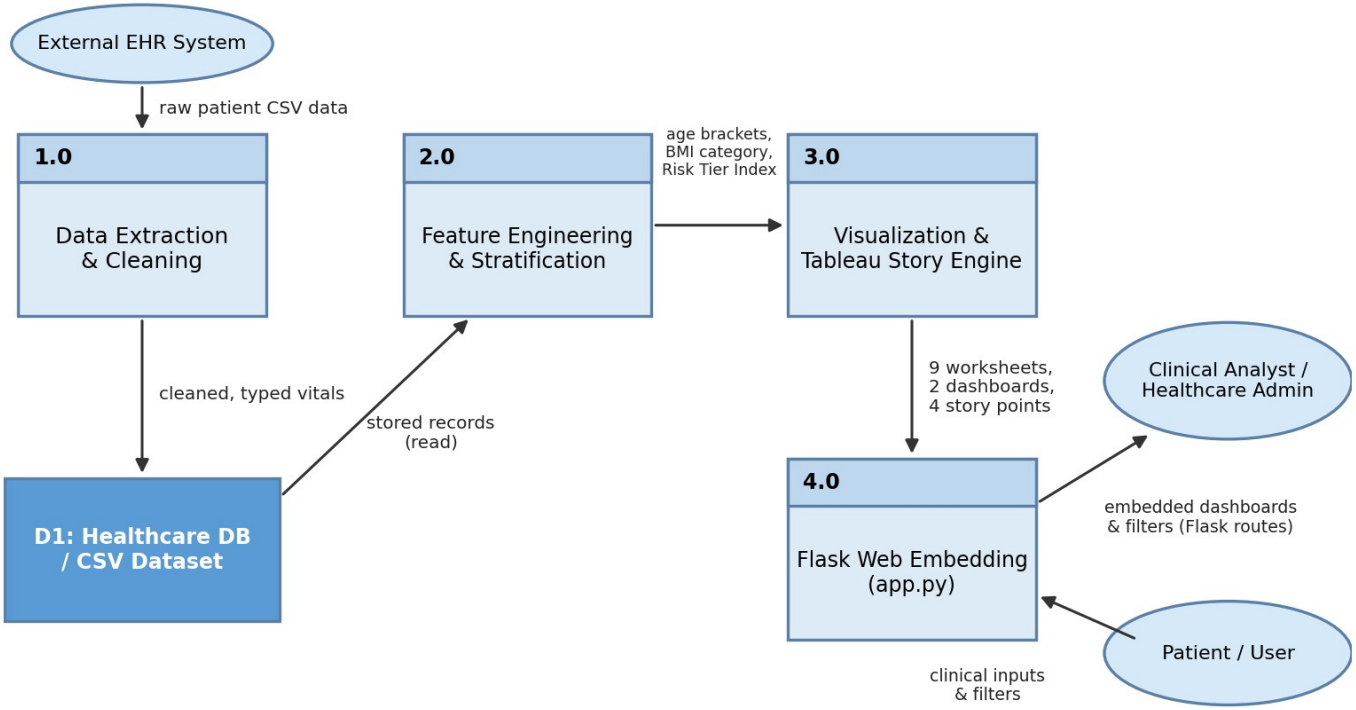


Table: DFD Level 0 elements.

Element	Type	Description
Patient / User	**External Entity**	Inputs clinical parameters and selects demographic filters.
Clinical Analyst / Healthcare Administrator	**External Entity**	Monitors population risk and reviews symptom burden.
External EHR System	**External Entity**	Outputs raw patient CSV data.
0.0 Health360 Analytics Engine & Web Portal	**Process**	Cleans and stratifies the patient data and serves interactive risk dashboards to users.
Healthcare Database / CSV Dataset	**Data Store**	smart_healthcare_dataset.csv – the 5,000-record patient dataset (vitals and lifestyle factors).

Table: DFD Level 1 processes.

Process	Name	Description
1.0	**Data Extraction & Cleaning**	Ingests raw CSV data, handles missing values, normalizes vital data types.
2.0	**Feature Engineering & Stratification**	Calculates Age Brackets (<20, 20-35, 36-60, 60+), WHO BMI Categories, and the Dynamic Risk Tier Index (High, Moderate, Low Risk).
3.0	**Visualization & Tableau Story Engine**	Renders 9 visual worksheets, 2 master dashboards, and 4 executive story points.
4.0	**Flask Web Embedding**	Embeds Tableau views into the HTML5 shell using Tableau JS API v2 and serves responses over Flask API routes (app.py).

User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Healthcare Administrator	Cohort Filtering	US-01	As a Healthcare Administrator, I want to filter patient records by Age Brackets (<20 to 60+), so that I can identify the early-onset hypertension baseline in younger patient cohorts.	Age bracket filter updates the KPI cards and charts for the selected cohort	High	Sprint-1
Clinical Risk Manager	Risk Stratification	US-02	As a Clinical Risk Manager, I want to view a Glucose vs. BMI scatter plot matrix color-coded by Dynamic Risk Index, so that I can stratify high-risk chronic disease clusters in real-time.	Scatter plot shows Glucose vs. BMI, colored by Dynamic Risk Index	High	Sprint-1
Chief Medical Officer	Symptom Burden Analysis	US-03	As a Chief Medical Officer, I want to review the Symptom Burden Heatmap across co-occurring conditions, so that I can allocate hospital resources and triage medical staff effectively.	Heatmap shows patient counts by primary condition and symptom burden (0-3)	High	Sprint-1
Data Analyst	Lifestyle Impact Analysis	US-04	As a Data Analyst, I want to analyze the compounding impacts of smoking, alcohol, and exercise on vitals, so that I can design targeted preventive lifestyle management programs.	Lifestyle Impact and Exercise vs. Vitals charts show risk index and blood pressure by lifestyle factor	Medium	Sprint-2
Primary Care Physician	Executive Story	US-05	As a Primary Care Physician, I want to navigate an interactive 4-scene Executive Story container, so that I can walk executive decision-makers through population trends cleanly.	All 4 story points can be opened in sequence from the story navigator	Medium	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Web Developer / System Admin	Web Embedding	US-06	As a Web Developer / System Admin, I want to access embedded Tableau analytics inside a Flask web application shell, so that clinicians no longer rely on external Tableau desktop clients.	Dashboards load in the Flask-served HTML5 page via Tableau JS API v2	High	Sprint-1

Figure 1: DFD Architecture & Data Flow Overview (Flask Web Shell – Health360 Analytics Platform)

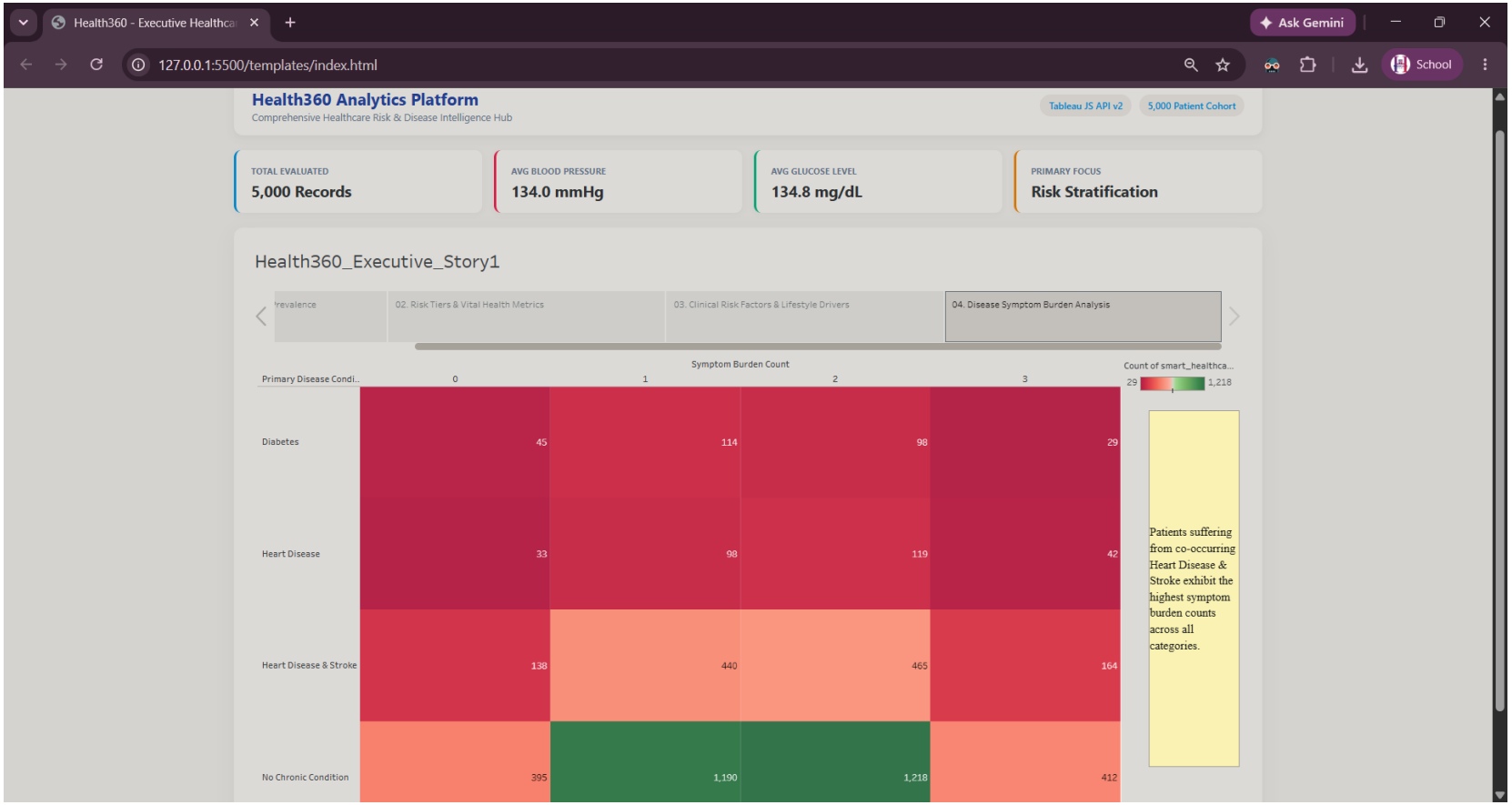


Figure 2: Executive Dashboard 1 – Population Overview & Vital Metrics

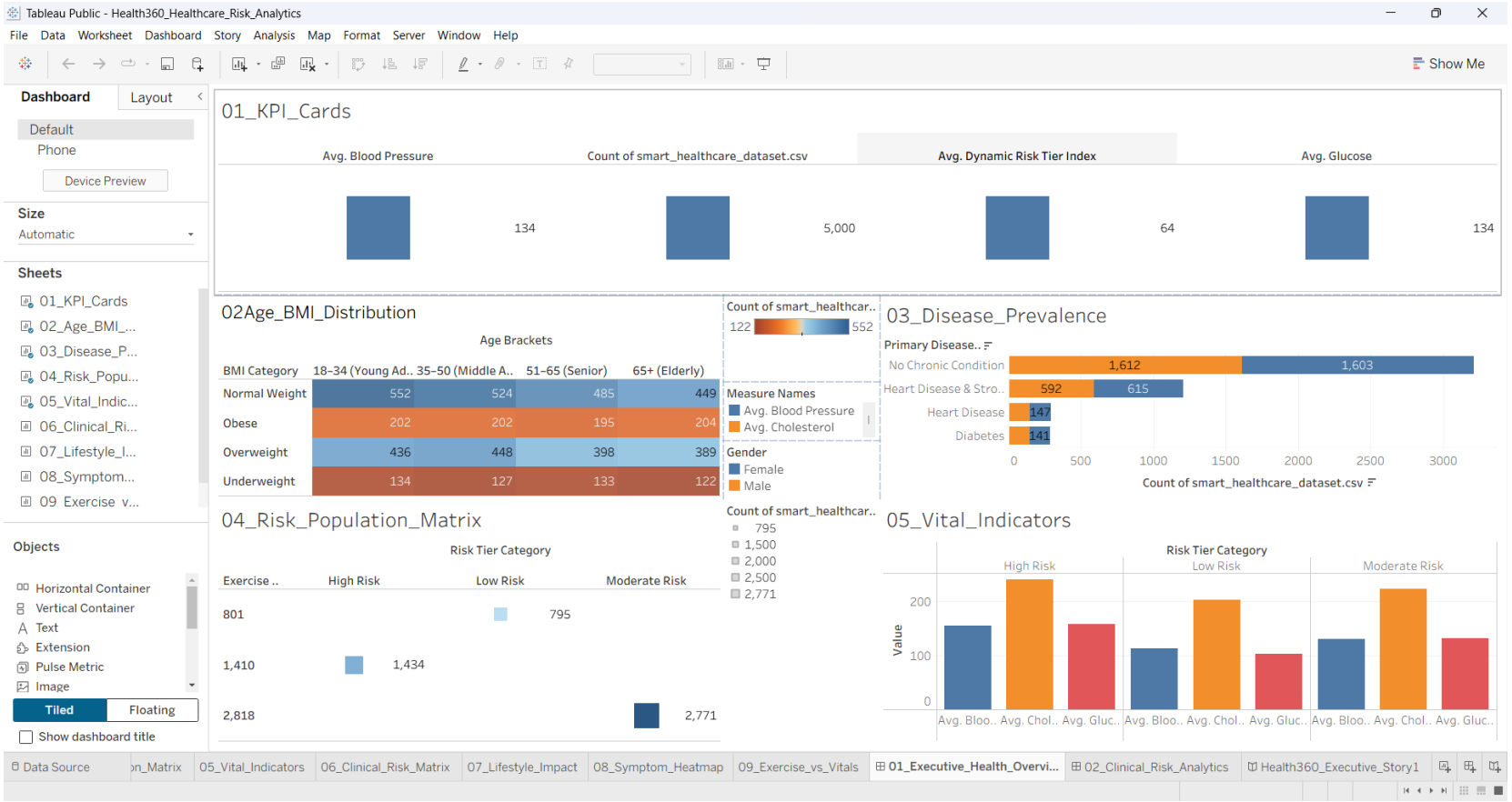


Figure 3: Clinical Risk Matrix & Symptom Burden Heatmap (Executive Dashboard 2)

